

Project Title: Enterprise Semantic Document Parser & RAG Knowledge Pipeline

Organization: Cloud Infrastructure & Platform Engineering Group

Date: March 2026 | Status: Architecture Review

Description:

A high-throughput distributed microservice architecture designed for automated parsing of unstructured PDF and DOCX technical reports. The platform incorporates sentence-transformers for 768-dimensional dense vector embeddings, FAISS vector indexing for similarity queries, and Celery asynchronous queues for high-concurrency background extraction.

Problem Statement:

Multiple internal product teams repeatedly build redundant document parsing logic and isolated vector search indexes, resulting in severe resource fragmentation, duplicate cloud costs, and high ingestion latency during quarterly vendor document intake.

Core Objectives:

1. Deliver standardized document chunking and metadata extraction across all corporate formats.
2. Accelerate semantic vector retrieval latency to under 15 milliseconds via persistent FAISS indexing.
3. Establish unified REST and gRPC endpoints for cross-team service reuse.

Technology Stack:

Programming Languages: Python, TypeScript, SQL

Frameworks & Libraries: FastAPI, React, Celery, LangChain

Databases & Storage: PostgreSQL, Redis, pgvector, FAISS

APIs & Cloud: OpenAI API, AWS S3, REST API, Docker, Kubernetes

AI / ML Tools: Sentence-Transformers, PyTorch, Vector Embeddings, RAG

Repository & Documentation:

GitHub Repository: <https://github.com/enterprise-org/rag-doc-pipeline>

Documentation: <https://docs.internal/architecture/rag-pipeline>